

Ethical Hacking

Devrim Celik
Hochschule Bonn-Rhein-Sieg
WS 2025

Formalities & Setup

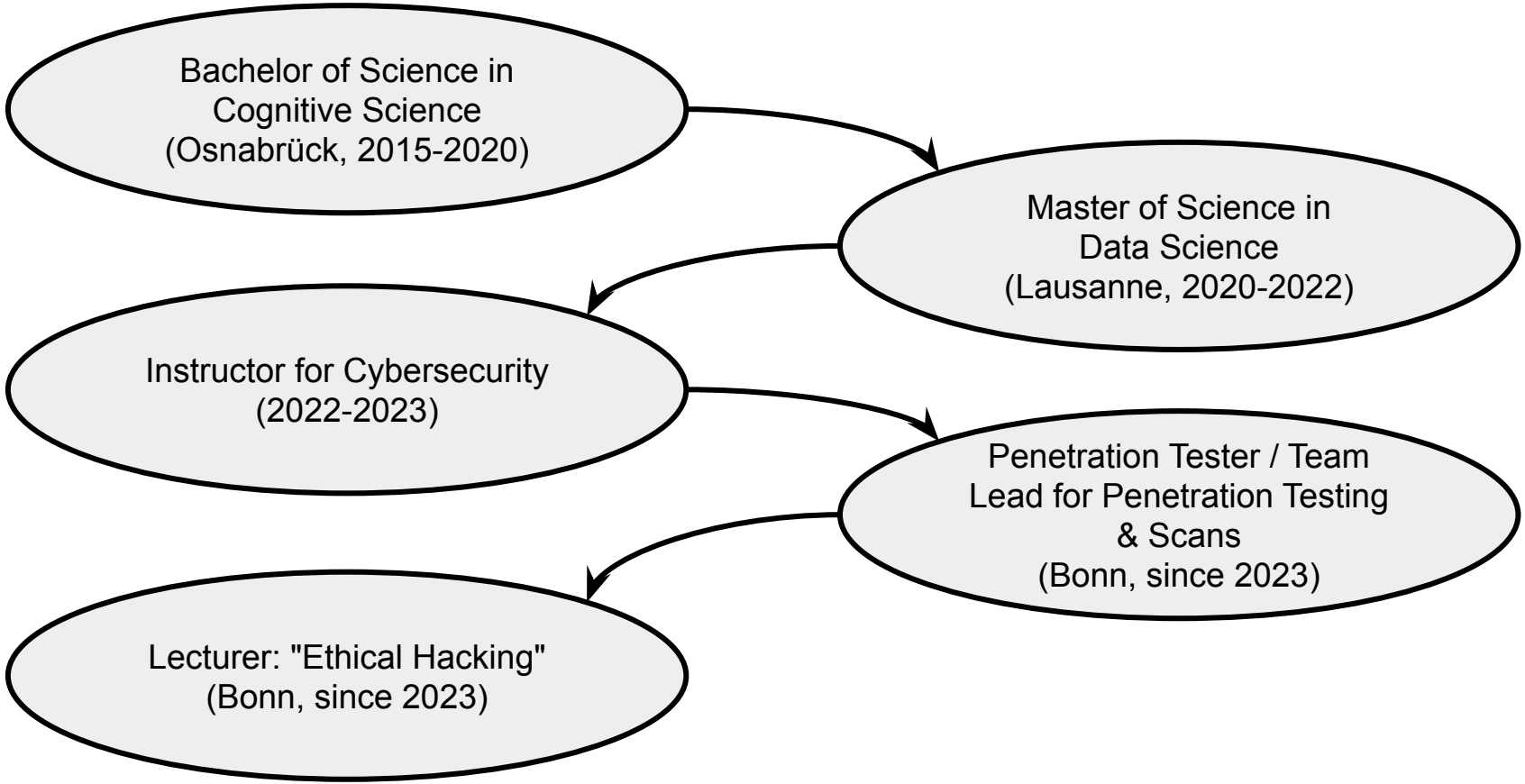
1	`whoami`
2	structure, content & timeline
3	communication channels
4	technical setup

Formalities & Setup

1	`whoami`
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`whoami`

- really enjoy teaching (me being here = my hobby)
  - my goal is to create the best learning experience i can for you
    - i will respect your opinion and your time
    - i'd like that you treat me the same way
  - call me by my first name


Bachelor of Science in
Cognitive Science
(Osnabrück, 2015-2020)
Master of Science in
Data Science
(Lausanne, 2020-2022)
Instructor for Cybersecurity
(2022-2023)
Penetration Tester / Team
Lead for Penetration Testing
& Scans
(Bonn, since 2023)
Lecturer: "Ethical Hacking"
(Bonn, since 2023)

why did you choose this lecture?
what are you expectation?

Formalities & Setup

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- this course is still in a very dynamic state, i.e., i still play around with things and change them
  - sometimes things will be a bit chaotic with me, thank you very much in advance for you patience
- course went through multiple iterations
  - first iteration was too technical
  - my goal for you: actually learning to hack linux system, boot2root
    - after this course: able to go onto common cybersecurity training platforms and be successful in pwning machines
  - ~50% of course will be me presenting and ~50% will you be hacking
    - first half of the course: mainly me presenting
    - second half the course: mainly you hacking
    - reason: i want you to be able to properly hack systems as soon as possible

Building Blocks

lecture

- theory
- me presenting
- fundamentals for proper hacking
- ask any questions

live demos

- me live hacking with commentary and Q&A
- to get a feel about what the thought-process is

hack sessions

- time during lecture time slots to work on the labs or Boot2Root machines
- i will be there for any questions you may have

labs

- exercise to apply theory
- designed to help you actually understand how to exploit vulnerabilities
- required for exam admission

TryHackMe boot2root

- THM is a platform for learning cybersecurity
- offers free systems for you to try and hack
- i will release a list with machines
- required for exam admission

custom boot2root

- middle of the semester, i will release a boot2root systems
- you will create a write-up
- required for exam admission

Building Blocks

exam admission requirements

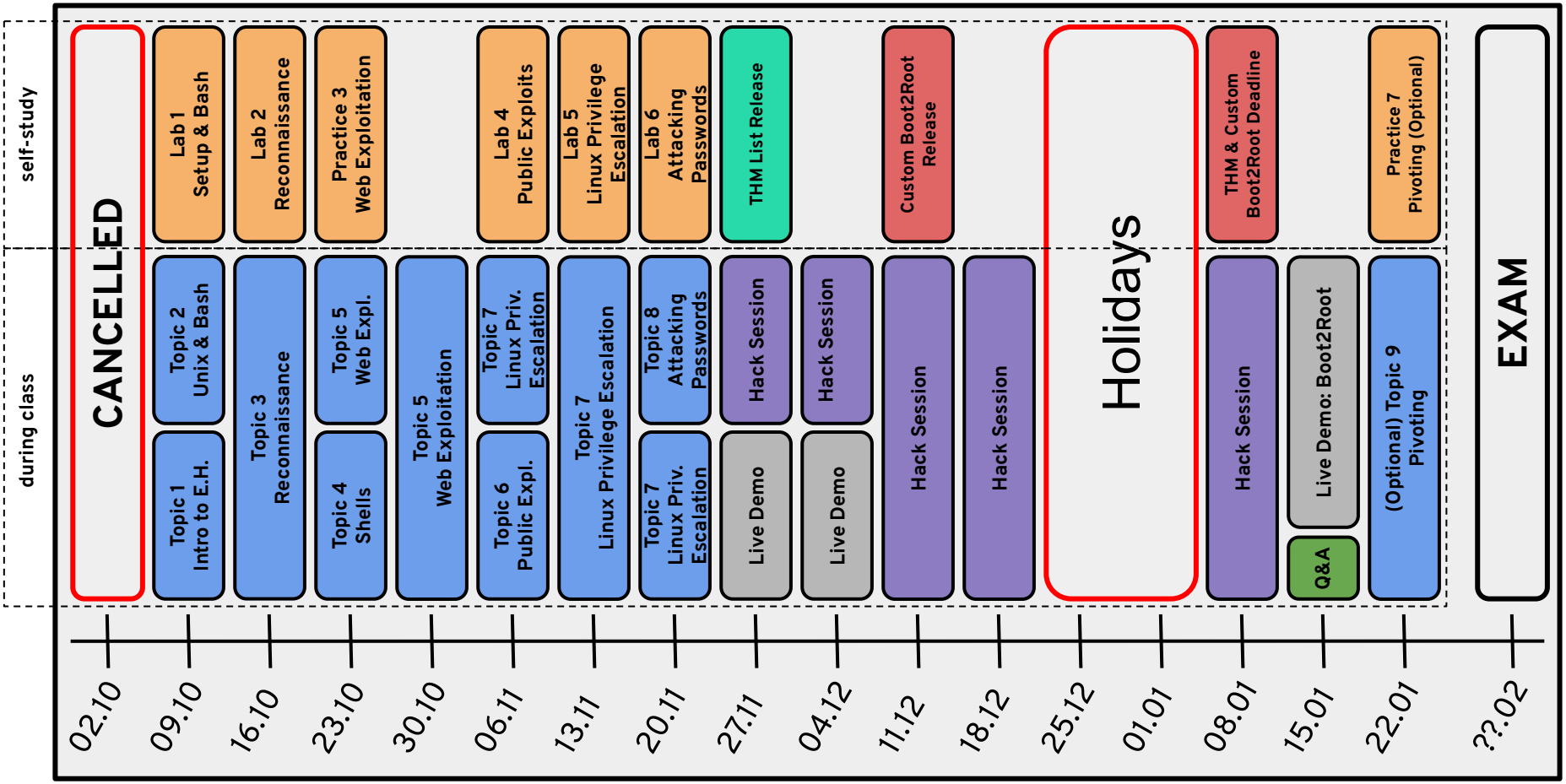
- **labs**
  - 6 released
  - 6 (short) write-ups required
  - deadline ⇒ 4 weeks after release
- **THM boot2root**
  - 5 released
  - 2 write-ups required
  - deadline ⇒ 08.01.26
- **custom boot2root**
  - 1 released
  - 1 write-up required
  - deadline ⇒ 08.01.26

Building Blocks

exam

- written
- 100% of the final grade
- will take place beginning of February (currently ⇒ 05.02.26)
- combination of
  - multiple choice questions
  - open questions
  - multi-step exploit riddles
- will test your
  - understanding of presented theory
  - exploiting-skills you learned about in the labs and boot2root machines
- i will try not to ask you anything that is unreasonable; i want to be fair
- will be standardized

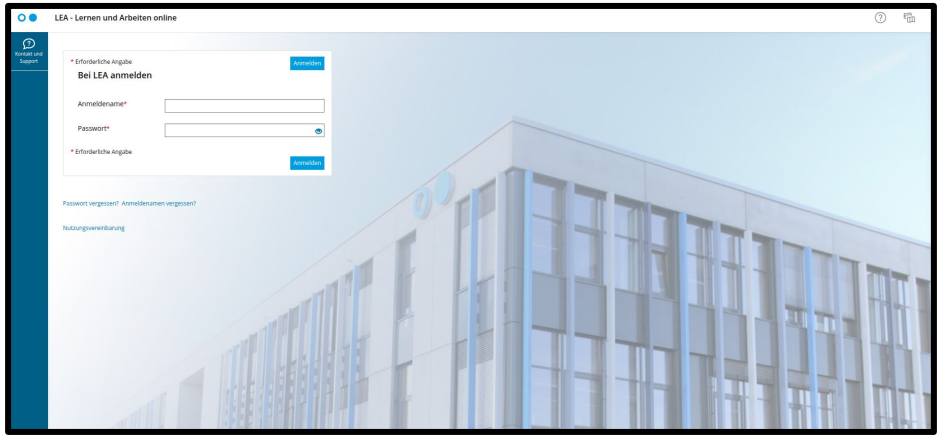
Timeline


self-study
during class
CANCELLED
Topic 1 Intro to E.H.
Lab 1 Setup & Bash
Topic 2 Unix & Bash
Lab 2 Reconnaissance
Topic 3 Reconnaissance
Practice 3 Web Exploitation
Topic 4 Shells
Topic 5 Web Expl.
Topic 5 Web Exploitation
Lab 4 Public Exploits
Topic 6 Public Expl.
Topic 7 Linux Priv. Escalation
Lab 5 Linux Privilege Escalation
Topic 7 Linux Privilege Escalation
Lab 6 Attacking Passwords
Topic 7 Linux Priv. Escalation
Topic 8 Attacking Passwords
THM List Release
Hack Session
Live Demo
Hack Session
Live Demo
Custom Boot2Root Release
Hack Session
Hack Session
Holidays
THM & Custom Boot2Root Deadline
Hack Session
Q&A
Live Demo: Boot2Root
Practice 7 Pivoting (Optional)
(Optional) Topic 9 Pivoting
EXAM
02.10
09.10
16.10
23.10
30.10
06.11
13.11
20.11
27.11
04.12
11.12
18.12
25.12
01.01
08.01
15.01
22.01
??.02

Formalities & Setup

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Communication Channel - LEA



- used for
  - sharing instruction documents, slides, labs, exercises with you
  - submitting your write ups

Communication Channel - Discord 1



- invite link: https://discord.gg/TQsjFDvE
- used for
  - asking and answering questions (lecture, labs, orga)
  - getting updates about the lecture from me
  - direct communication with me

Communication Channel - Discord 2



- pls rename yourself within the context of the channel
  - Devrim Celik → Devrim C.
- optional, but in case you care, you can deactivate “show activity” in your privacy settings in Discord

Communication Channel - Discord 2

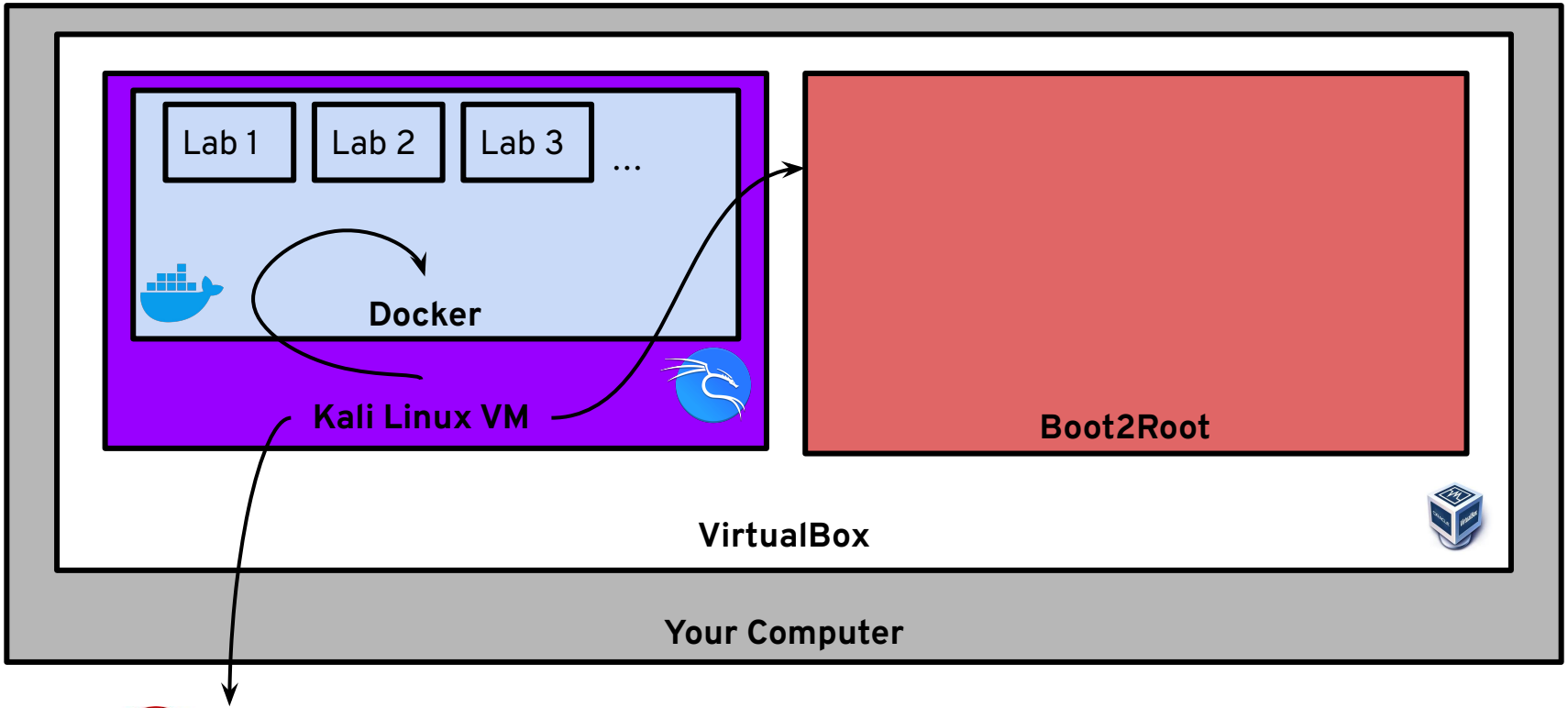


- education@dcelik.net
- used for
  - direct communication with me about official/important topics

Formalities & Setup

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Technical Setup



Try Hack Me

guide for installation and setup
was already uploaded on LEA

Technical Setup

